

Review retrospective:
*the 1D anharmonic rescaling identity, and four pieces of dead
code*

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Abstract

A soundness review of `Laplace/OneD/Rescaling.lean`: the rescaling identity $tL(u/\sqrt{\lambda t}) = u^2/2 + s_t(u)$ for the anharmonic potential, with $s_t(u) = \frac{\alpha}{6\lambda^{3/2}} \frac{u^3}{\sqrt{t}} + \frac{\gamma}{24\lambda^2} \frac{u^4}{t}$, plus the coercivity-driven Gaussian-decay bounds and the polynomial bound $s_t(u)^2 \leq 2(A^2 + B^2)(u^6 + u^8)/t$. Result: **a clean fidelity pass with four dead-code removals** — a dead `open Real` and three unused local hypotheses, all rebuild-arbitrated with the `unusedVariables` linter as final witness.

1 Setting

This is the rescaling layer of the 1D anharmonic Laplace expansion (primer Stage 2): the substitution $x = u/\sqrt{\lambda t}$ normalises the harmonic part of tL to $u^2/2$ and packages the rest as the remainder $s_t(u)$, with cubic and quartic coefficients $A = \alpha/(6\lambda^{3/2})$, $B = \gamma/(24\lambda^2)$. It is the 1D original that the multivariate QuadraticApprox mirrors, and the first review into the non-cumulant 1D layer.

2 What was reviewed

Three definitions (`cubicScale`, `quarticScale`, `rescaledPerturbation`), the rescaling identity and its `symm` rephrasing, two Gaussian-decay theorems (`rescaled_boltzmann_decay` and `rescaled_max_decay`), and the polynomial bound `rescaledPerturbation_sq_le` (plus one private $(a+b)^2 \leq 2a^2 + 2b^2$ helper).

3 Fidelity / soundness findings

Sound; the mathematics is faithful. The load-bearing check was the rescaling algebra, and it is correct: with $L = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$ and $x = u/\sqrt{\lambda t}$, the cubic term gives $t \cdot \frac{\alpha}{6}x^3 = \frac{\alpha}{6}u^3 \frac{t}{(\lambda t)^{3/2}} = \frac{\alpha}{6\lambda^{3/2}} \frac{u^3}{\sqrt{t}}$ (the key step $t/(\lambda t)^{3/2} = 1/(\lambda^{3/2}\sqrt{t})$), and the quartic term gives $\frac{\gamma}{24\lambda^2} \frac{u^4}{t}$ — exactly the stated coefficients. The decay bounds correctly thread coercivity through the rescaling ($c_0 = c/\lambda$, then $\min(1/2, c)$), and the polynomial bound has the right exponents $u^6 + u^8$ and the right $t \geq 1$ usage $1/t^2 \leq 1/t$.

One benign note, recorded but **not** escalated and **not** acted on: the two decay theorems carry a hypothesis $\gamma > 0$ that is already forced by $\lambda > 0$ and the discriminant $\alpha^2 < 3\lambda\gamma$, so the Lean statements are very slightly stronger than the minimal claim. Removing it is out of a review tide's

scope — dropping a public hypothesis changes the signature, and besides, the proof genuinely *uses* $\gamma > 0$ to invoke `anharmonic_coercive` (whose own signature demands it), so weakening here would cascade an API change through the dependency. Harmless and left as-is.

4 Simplifications applied

Four dead-code removals. A dead open `Real` (every `Real.sqrt/Real.exp/Real.sq_sqrt` is fully qualified; the only $\sqrt{}$ symbols are in comments), and three unused local hypotheses: `hsqrt_lamt_pos` in the rescaling identity (the power computations use `Real.sq_sqrt hlamt.le` directly), and `hsqrt_t_pos/hsqrt_t_ne` in the polynomial bound (`hsqrt_t_sq` comes straight from `ht_pos.le`; the surviving `ht_ne` is what `field_simp` actually consumes). All four removed in one worktree and **rebuild-arbitrated**: the file rebuilt green (2699 jobs) with *no* remaining `unusedVariables` warning and all nine declarations byte-identical. This last point is the discipline that matters — the seabed’s standing gotcha is that “dead” locals sometimes feed a later tactic implicitly, so a reviewer’s unused-flag is only a candidate; the build plus the linter are the arbiter, and here all four were confirmed genuinely dead.

GPT’s six other proposals — inline the `field_simp` nonzero facts, trim the `nlinarith` hint list, inline the one-use private helper, inline `ht_ne` (which is in fact used), shorten the `hLHS` rewrite, and collapse the `hkey/hgoal` staging — are tactic-golf or inlining and were declined.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A reviewer’s “unused local” is a candidate, not a verdict.** Three of the four removals here were local hypotheses GPT called unused; the honest confirmation was the rebuild with the `unusedVariables` linter clean afterwards, not the reviewer’s say-so. (The seabed has a documented case where a “dead” local was load-bearing for a downstream `linarith`.) Remove the candidates, rebuild, and let the linter’s silence be the proof.
- **A redundant hypothesis that the proof actually uses is not removable as a simplification.** The $\gamma > 0$ on the decay theorems is mathematically implied by the other hypotheses, but it is passed to `anharmonic_coercive`, so removing it is an API change cascading through a dependency — a forward tide at most, never a review-tide signature-preserving edit.

7 Follow-ups

- The 1D non-cumulant layer is large and now barely opened. Tractable never-reviewed neighbours include `GaussianMoments.lean`, `Localisation.lean`, `Harmonic.lean`, and the J-function files; the `Multi/Covariance*` trio (3998–19611 lines) still awaits a sectioned approach.